

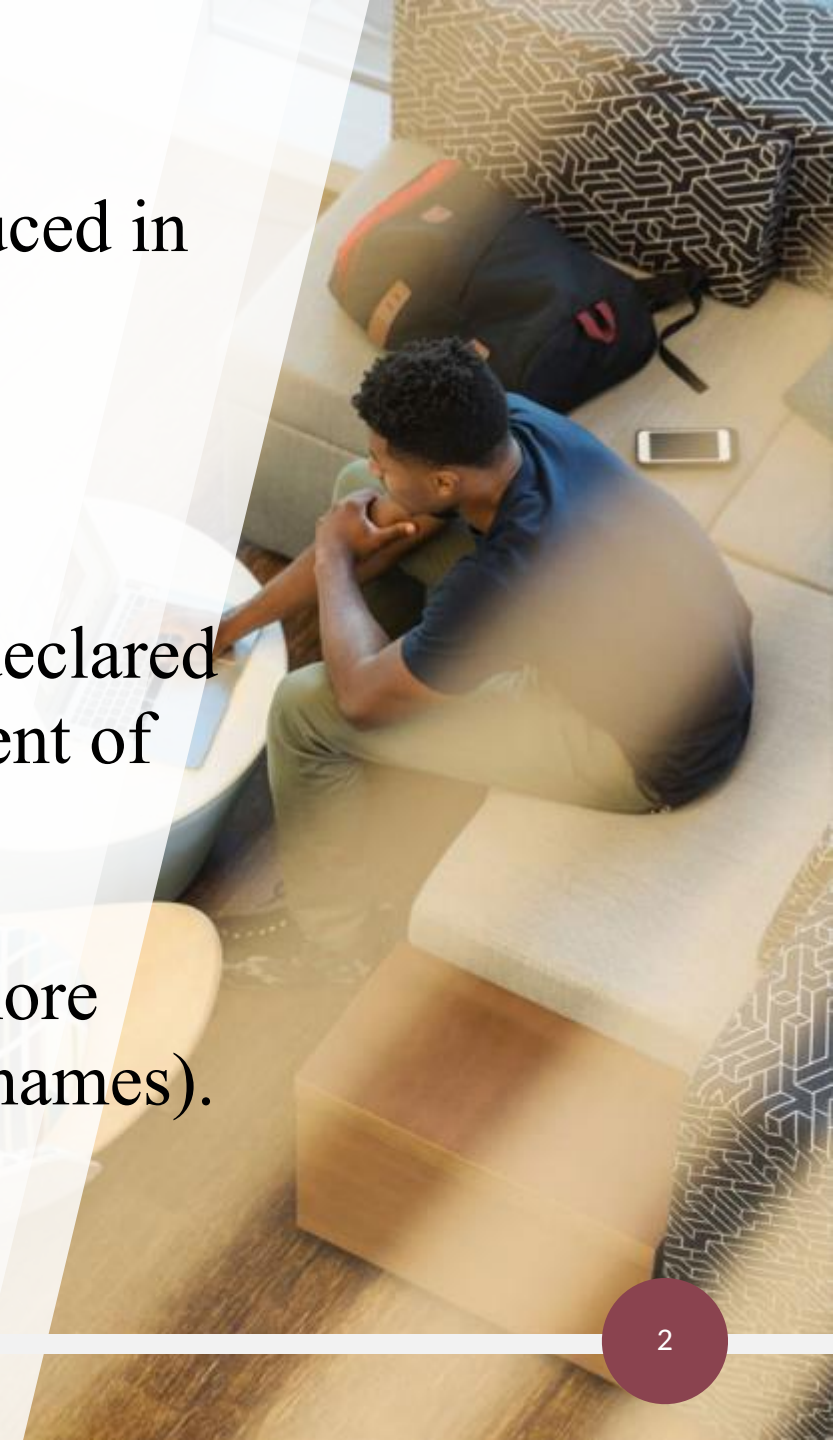
A background image of four students in a library. A young woman with long dark hair is on the left, smiling. Next to her is a young man with dark hair, also smiling. To his right is a young woman with glasses and dark hair, looking towards the right. On the far right is a young man with dark hair, seen from the back/side, looking at a laptop. They are all sitting at a table with books and a laptop. The background is filled with bookshelves.

Java 25

Unnamed Variables and Patterns

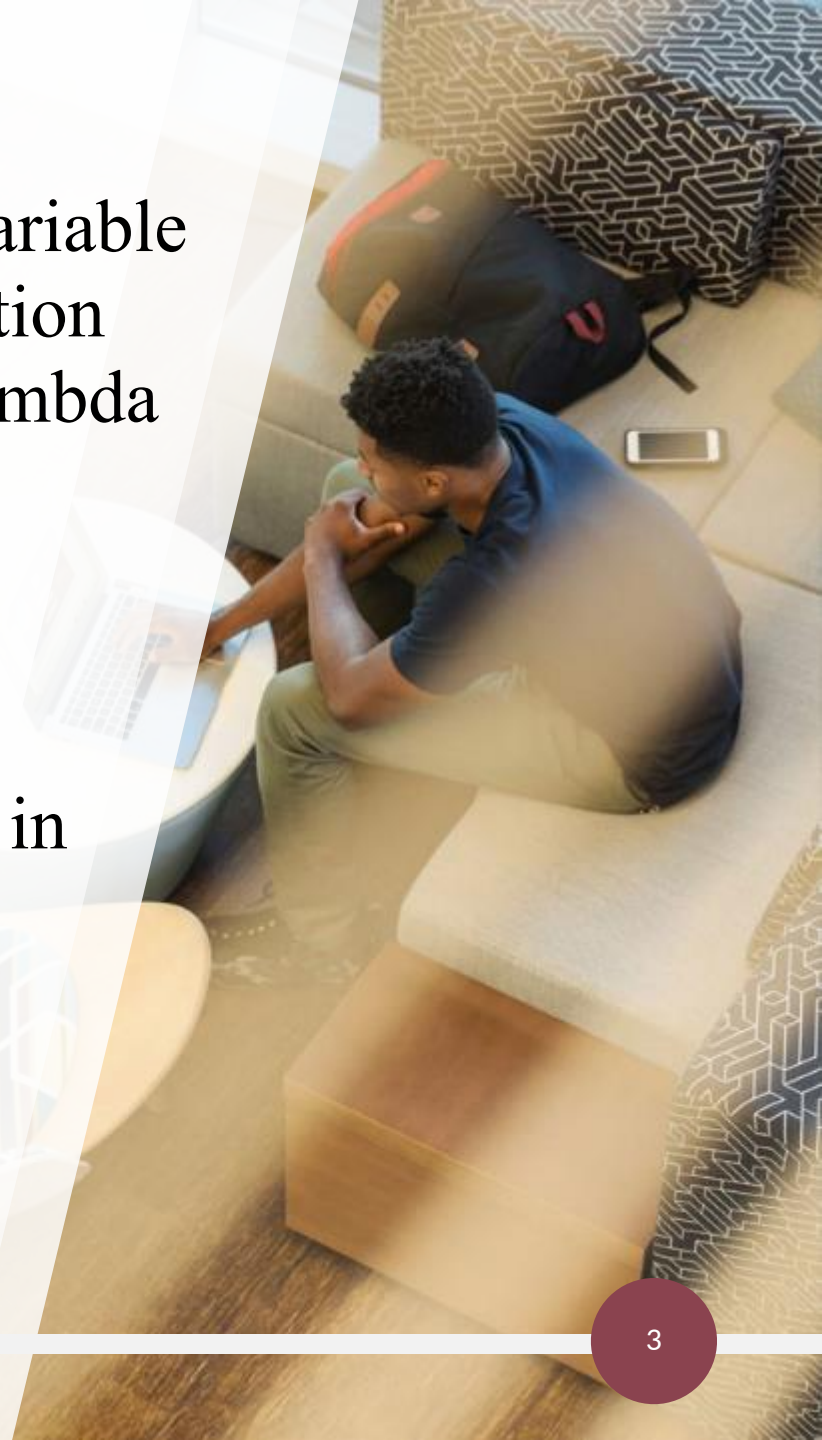
Background and Motivation

- Unnamed variables and unnamed patterns were introduced in Java 21 and are now finalised in Java 22.
- Both are denoted with the underscore character, `_`.
- The goal here is that in situations where variables are declared with no intention to use them, we can now make the intent of *non-use* explicit.
- This also makes the code cleaner, more readable and more expressive (as we are eliminating unnecessary variable names).



Unnamed Variables

- Unnamed variables apply to local variables e.g. local variable declarations, headers in basic/enhanced *for* loops, exception parameters in *catch* clauses and lambda parameters in lambda expressions.
- Unnamed variables do not apply to instance variables.
- Declaring a variable that is never used is very common in code whose side-effect is very important.



Unnamed Variables : try-with-resources, *catch*

```
7  ▶ public class UnnamedVariableTry {
8  ▶     public static void main(String[] args) {
9      try (BufferedReader in =    // 'in' not used
10         new BufferedReader(new FileReader( fileName: "example.txt"))) {
11         System.out.println("File opened successfully!");
12     } catch (IOException ioe) { // 'ioe' not used
13         System.out.println("Error reading the file.");
14     }
15     try (BufferedReader _ =    // 'in' not even named
16         new BufferedReader(new FileReader( fileName: "example.txt"))) {
17         System.out.println("File opened successfully!");
18     } catch (IOException _) {  // 'ioe' not named
19         System.out.println("Error reading the file.");
20     }
21 }
22 }
```

Unnamed Patterns

- Unnamed patterns enable us to ignore specific fields when using pattern matching.
- Particularly useful in record deconstruction and switch expressions.

